



National  
Qualifications  
2022

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**2022 Mathematics**

**Higher**

**Paper 1 (Non-calculator)**

**Finalised Marking Instructions**

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## General marking principles for Higher Mathematics

*Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.*

*For each question, the marking instructions are generally in two sections:*

- *generic scheme – this indicates why each mark is awarded*
- *illustrative scheme – this covers methods which are commonly seen throughout the marking*

*In general, you should use the illustrative scheme. Only use the generic scheme where a candidate has used a method not covered in the illustrative scheme.*

- Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.
- If you are uncertain how to assess a specific candidate response because it is not covered by the general marking principles or the detailed marking instructions, you must seek guidance from your team leader.
- One mark is available for each •. There are no half marks.
- If a candidate's response contains an error, all working subsequent to this error must still be marked. Only award marks if the level of difficulty in their working is similar to the level of difficulty in the illustrative scheme.
- Only award full marks where the solution contains appropriate working. A correct answer with no working receives no mark, unless specifically mentioned in the marking instructions.
- Candidates may use any mathematically correct method to answer questions, except in cases where a particular method is specified or excluded.
- If an error is trivial, casual or insignificant, for example  $6 \times 6 = 12$ , candidates lose the opportunity to gain a mark, except for instances such as the second example in point (h) below.
- If a candidate makes a transcription error (question paper to script or within script), they lose the opportunity to gain the next process mark, for example

This is a transcription error and so the mark is not awarded.

This is no longer a solution of a quadratic equation, so the mark is not awarded.

$$x^2 + 5x + 7 = 9x + 4$$

$$x - 4x + 3 = 0$$

$$x = 1$$

The following example is an exception to the above

This error is not treated as a transcription error, as the candidate deals with the intended quadratic equation. The candidate has been given the benefit of the doubt and all marks awarded.

$$x^2 + 5x + 7 = 9x + 4$$

$$x - 4x + 3 = 0$$

$$(x - 3)(x - 1) = 0$$

$$x = 1 \text{ or } 3$$

(i) **Horizontal/vertical marking**

If a question results in two pairs of solutions, apply the following technique, but only if indicated in the detailed marking instructions for the question.

Example:

$$\begin{array}{cc} \bullet^5 & \bullet^6 \\ \bullet^5 & x = 2 \quad x = -4 \\ \bullet^6 & y = 5 \quad y = -7 \end{array}$$

Horizontal:  $\bullet^5 x = 2$  and  $x = -4$       Vertical:  $\bullet^5 x = 2$  and  $y = 5$   
 $\bullet^6 y = 5$  and  $y = -7$        $\bullet^6 x = -4$  and  $y = -7$

You must choose whichever method benefits the candidate, **not** a combination of both.

(j) In final answers, candidates should simplify numerical values as far as possible unless specifically mentioned in the detailed marking instruction. For example

$$\begin{array}{ll} \frac{15}{12} \text{ must be simplified to } \frac{5}{4} \text{ or } 1\frac{1}{4} & \frac{43}{1} \text{ must be simplified to } 43 \\ \frac{15}{0.3} \text{ must be simplified to } 50 & \frac{4}{\cancel{5}} \text{ must be simplified to } \frac{4}{15} \\ \sqrt{64} \text{ must be simplified to } 8^* & \end{array}$$

\*The square root of perfect squares up to and including 100 must be known.

- (k) Commonly Observed Responses (COR) are shown in the marking instructions to help mark common and/or non-routine solutions. CORs may also be used as a guide when marking similar non-routine candidate responses.
- (l) Do not penalise candidates for any of the following, unless specifically mentioned in the detailed marking instructions:

- working subsequent to a correct answer
- correct working in the wrong part of a question
- legitimate variations in numerical answers/algebraic expressions, for example angles in degrees rounded to nearest degree
- omission of units
- bad form (bad form only becomes bad form if subsequent working is correct), for example

$$(x^3 + 2x^2 + 3x + 2)(2x + 1) \text{ written as}$$

$$(x^3 + 2x^2 + 3x + 2) \times 2x + 1$$

$$= 2x^4 + 5x^3 + 8x^2 + 7x + 2$$

gains full credit

- repeated error within a question, but not between questions or papers

(m) In any 'Show that...' question, where candidates have to arrive at a required result, the last mark is not awarded as a follow-through from a previous error, unless specified in the detailed marking instructions.

(n) You must check all working carefully, even where a fundamental misunderstanding is apparent early in a candidate's response. You may still be able to award marks later in the question so you must refer continually to the marking instructions. The appearance of the correct answer does not necessarily indicate that you can award all the available marks to a candidate.

- (o) You should mark legible scored-out working that has not been replaced. However, if the scored-out working has been replaced, you must only mark the replacement working.
- (p) If candidates make multiple attempts using the same strategy and do not identify their final answer, mark all attempts and award the lowest mark. If candidates try different valid strategies, apply the above rule to attempts within each strategy and then award the highest mark.

For example:

|                                                                    |                                                                    |
|--------------------------------------------------------------------|--------------------------------------------------------------------|
| Strategy 1 attempt 1 is worth 3 marks.                             | Strategy 2 attempt 1 is worth 1 mark.                              |
| Strategy 1 attempt 2 is worth 4 marks.                             | Strategy 2 attempt 2 is worth 5 marks.                             |
| From the attempts using strategy 1, the resultant mark would be 3. | From the attempts using strategy 2, the resultant mark would be 1. |

In this case, award 3 marks.

## Marking Instructions for each question

| Question                                                                                                                                                                                                                                                                                                                                                              |  |  | Generic scheme                                                                                                                                                                  | Illustrative scheme                                                                                                                                                                        | Max mark |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1.                                                                                                                                                                                                                                                                                                                                                                    |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> state gradient</li> <li>•<sup>2</sup> state perpendicular gradient</li> <li>•<sup>3</sup> find equation of line</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>-\frac{5}{2}</math></li> <li>•<sup>2</sup> <math>\frac{2}{5}</math></li> <li>•<sup>3</sup> <math>5y = 2x + 32</math></li> </ul> | 3        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                         |  |  |                                                                                                                                                                                 |                                                                                                                                                                                            |          |
| 1. At • <sup>1</sup> , ignore any errors in processing the constant term.<br>2. At • <sup>1</sup> and • <sup>2</sup> , ignore the appearance of 'x'.<br>3. • <sup>3</sup> is only available as a consequence of using a perpendicular gradient.<br>4. At • <sup>3</sup> , accept any arrangement of a candidate's equation where constant terms have been simplified. |  |  |                                                                                                                                                                                 |                                                                                                                                                                                            |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                   |  |  |                                                                                                                                                                                 |                                                                                                                                                                                            |          |
| <b>Candidate A</b><br>A perpendicular gradient has been clearly stated<br>$5x + 2y = 7$<br><br>$m_{\perp} = \frac{2}{5}$<br>$5y = 2x + 32$                                                                                                                                                                                                                            |  |  | • <sup>1</sup> ✓ • <sup>2</sup> ✓<br>• <sup>3</sup> ✓                                                                                                                           | <b>Candidate B</b><br>No communication for perpendicular gradient<br>$5x + 2y = 7$<br>$y = -\frac{5}{2}x + \frac{7}{2}$<br>$m = \frac{2}{5}$<br>$5y = 2x + 32$                             |          |
| <b>Candidate C</b><br>$m = 5$<br>$m_{\perp} = -\frac{1}{5}$<br>$x + 5y = 29$                                                                                                                                                                                                                                                                                          |  |  | • <sup>1</sup> ✗<br>• <sup>2</sup> ✓ 1<br>• <sup>3</sup> ✓ 1                                                                                                                    | • <sup>1</sup> ^ • <sup>2</sup> ✓ 1<br>• <sup>3</sup> ✓ 1                                                                                                                                  |          |

| Question                                                                                                                                            |  |  | Generic scheme                                                                                                                                                                                                                | Illustrative scheme                                                                                                                                                       | Max mark |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2.                                                                                                                                                  |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> apply <math>m \log_n x = \log_n x^m</math></li> <li>•<sup>2</sup> apply <math>\log_n x - \log_n y = \log_n \frac{x}{y}</math></li> <li>•<sup>3</sup> evaluate</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\log_3 6^2</math></li> <li>•<sup>2</sup> <math>\log_3 \frac{6^2}{4}</math></li> <li>•<sup>3</sup> 2</li> </ul> | 3        |
| <b>Notes:</b>                                                                                                                                       |  |  |                                                                                                                                                                                                                               |                                                                                                                                                                           |          |
| 1. Do not penalise the omission of the base of the logarithm at • <sup>1</sup> or • <sup>2</sup> .<br>2. Correct answer with no working, award 0/3. |  |  |                                                                                                                                                                                                                               |                                                                                                                                                                           |          |
| <b>Commonly Observed Responses:</b>                                                                                                                 |  |  |                                                                                                                                                                                                                               |                                                                                                                                                                           |          |
| <b>Candidate A - introducing a variable</b><br>$\log_3 9$ • <sup>1</sup> ✓ • <sup>2</sup> ✓<br>$3^x = 9$<br>$x = 2$ • <sup>3</sup> ✓                |  |  | <b>Candidate B</b><br>$2 \log_3 \left( \frac{6}{4} \right)$ • <sup>2</sup> ✗<br>$\log_3 \left( \frac{6}{4} \right)^2$ • <sup>1</sup> <span style="border: 1px solid red; padding: 2px;">✓ 1</span> • <sup>3</sup> ^           |                                                                                                                                                                           |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  | Generic scheme                                                                                                                                                                                                                                                             | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                  | Max mark |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  | <p><b>Method 1</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> equate composite function to <math>x</math></li> <li>•<sup>2</sup> write <math>h(h^{-1}(x))</math> in terms of <math>h^{-1}(x)</math></li> <li>•<sup>3</sup> state inverse function</li> </ul> | <p><b>Method 1</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>h(h^{-1}(x)) = x</math></li> <li>•<sup>2</sup> <math>4 + \frac{1}{3}h^{-1}(x) = x</math></li> <li>•<sup>3</sup> <math>h^{-1}(x) = 3(x - 4)</math></li> </ul>                                                                                                                       | 3        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  | <p><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> write as <math>y = h(x)</math> and start to rearrange</li> <li>•<sup>2</sup> express <math>x</math> in terms of <math>y</math></li> <li>•<sup>3</sup> state inverse function</li> </ul>        | <p><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>y = h(x) \Rightarrow x = h^{-1}(y)</math><br/><math>y - 4 = \frac{1}{3}x</math> or <math>3y = 12 + x</math></li> <li>•<sup>2</sup> <math>x = 3(y - 4)</math></li> <li>•<sup>3</sup> <math>h^{-1}(y) = 3(y - 4)</math><br/><math>\Rightarrow h^{-1}(x) = 3(x - 4)</math></li> </ul> |          |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |  |                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                      |          |
| <p>1. In Method 1, accept <math>4 + \frac{1}{3}h^{-1}(x) = x</math> for •<sup>1</sup> and •<sup>2</sup>.</p> <p>2. In Method 2, accept '<math>y - 4 = \frac{1}{3}x</math>' without reference to <math>y = h(x) \Rightarrow x = h^{-1}(y)</math> at •<sup>1</sup>.</p> <p>3. In Method 2, accept <math>h^{-1}(x) = 3(x - 4)</math> without reference to <math>h^{-1}(y)</math> at •<sup>3</sup>.</p> <p>4. In Method 2, beware of candidates with working where each line is not mathematically equivalent. See Candidates A and B for example.</p> <p>5. At •<sup>3</sup> stage, accept <math>h^{-1}</math> written in terms of any dummy variable eg <math>h^{-1}(y) = 3(y - 4)</math>.</p> <p>6. <math>y = 3(x - 4)</math> does not gain •<sup>3</sup>.</p> <p>7. <math>h^{-1}(x) = 3(x - 4)</math> with no working gains 3/3.</p> |  |  |                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                      |          |

| Question                                                                                                                                | Generic scheme                                                      | Illustrative scheme                                                                                                                                                                       | Max mark                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 3.                                                                                                                                      | (continued)                                                         |                                                                                                                                                                                           |                                                                                            |
| Commonly Observed Responses:                                                                                                            |                                                                     |                                                                                                                                                                                           |                                                                                            |
| <b>Candidate A</b><br>$h(x) = 4 + \frac{1}{3}x$<br>$y = 4 + \frac{1}{3}x$<br>$x = 3(y - 4)$<br>$y = 3(x - 4)$<br>$h^{-1}(x) = 3(x - 4)$ | $\bullet^1 \checkmark$ $\bullet^2 \checkmark$<br>$\bullet^3 \times$ | <b>Candidate B</b><br>$h(x) = 4 + \frac{1}{3}x$<br>$y = 4 + \frac{1}{3}x$<br>$x = 4 + \frac{1}{3}y$<br>$y = 3(x - 4)$<br>$h^{-1}(x) = 3(x - 4)$                                           | $\bullet^1 \times$<br>$\bullet^2 \boxed{\checkmark 1}$<br>$\bullet^3 \boxed{\checkmark 1}$ |
| <b>Candidate C - BEWARE</b><br>$h' = \dots$                                                                                             | $\bullet^3 \times$                                                  | <b>Candidate D</b><br>$x \rightarrow x \div 3 \rightarrow x \div 3 + 4 = h(x)$<br>$\div 3 \rightarrow +4$<br>$\therefore -4 \rightarrow \times 3$<br>$3(x - 4)$<br>$h^{-1}(x) = 3(x - 4)$ | $\bullet^1 \checkmark$<br>$\bullet^2 \checkmark$<br>$\bullet^3 \checkmark$                 |



| Question                                                                                                                                                                          |  |  | Generic scheme                                                                                                                                                                                             | Illustrative scheme                                                                                                                                                                                                                                                 | Max mark |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4.                                                                                                                                                                                |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> express first term in differentiable form</li> <li>•<sup>2</sup> differentiate first term</li> <li>•<sup>3</sup> differentiate second term</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>y = x^{\frac{3}{2}} \dots</math> stated or implied by •<sup>2</sup></li> <li>•<sup>2</sup> <math>\frac{3}{2} x^{\frac{1}{2}} \dots</math></li> <li>•<sup>3</sup> <math>\dots + 2x^{-2}</math></li> </ul> | 3        |
| <b>Notes:</b>                                                                                                                                                                     |  |  |                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                     |          |
| 1. • <sup>2</sup> is only available for differentiating a term with a fractional index.<br>2. Where candidates attempt to integrate throughout, only • <sup>1</sup> is available. |  |  |                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                     |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                               |  |  |                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                     |          |
| Candidate A - differentiating over two lines                                                                                                                                      |  |  |                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                     |          |
| $\text{y} = x^{\frac{3}{2}} + 2x^{-2}$                                                                                                                                            |  |  | • <sup>1</sup> ✓                                                                                                                                                                                           |                                                                                                                                                                                                                                                                     |          |
| $\text{y} = \frac{3}{2} x^{\frac{1}{2}} + 2x^{-2}$                                                                                                                                |  |  | • <sup>2</sup> ✓ • <sup>3</sup> ✗                                                                                                                                                                          |                                                                                                                                                                                                                                                                     |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  | Generic scheme                                                                                                                                                                     | Illustrative scheme                                                                                                                                                                                                                                                                                            | Max mark |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  | <ul style="list-style-type: none"><li>•<sup>1</sup> use <math>m = \tan \theta</math></li><li>•<sup>2</sup> evaluate exact value</li><li>•<sup>3</sup> determine equation</li></ul> | <ul style="list-style-type: none"><li>•<sup>1</sup> <math>m = \tan \frac{\pi}{6}</math> or <math>m = \tan 30^\circ</math></li><li>•<sup>2</sup> <math>\frac{1}{\sqrt{3}}</math></li><li>•<sup>3</sup> eg <math>y\sqrt{3} = x + 2</math> or <math>y = \frac{1}{\sqrt{3}}x + \frac{2}{\sqrt{3}}</math></li></ul> | 3        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |  |                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                |          |
| <p>1. Do not award •<sup>1</sup> for <math>m = \tan^{-1} \frac{\pi}{6}</math>. However •<sup>2</sup> and •<sup>3</sup> are still available. Where candidates state <math>m = \tan^{-1} \frac{\pi}{3}</math> only •<sup>3</sup> is available.</p> <p>2. Where candidates make no reference to a trigonometric ratio or use an incorrect trigonometric ratio, •<sup>1</sup> and •<sup>2</sup> are unavailable.</p> <p>3. •<sup>3</sup> is only available as a consequence of attempting to use a tan ratio. See Candidate F</p> <p>4. Accept <math>y = \frac{1}{\sqrt{3}}(x + 2)</math> for •<sup>3</sup>, but do not accept <math>y - 0 = \frac{1}{\sqrt{3}}(x + 2)</math>.</p> |  |  |                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                |          |
| <b>Candidate A</b><br>$m = \tan \frac{\pi}{3}$<br>$m = \sqrt{3}$<br>$y = \sqrt{3}x + 2\sqrt{3}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  | <ul style="list-style-type: none"><li>•<sup>1</sup> ✗</li><li>•<sup>2</sup> ✓ 1</li><li>•<sup>3</sup> ✓ 1</li></ul>                                                                | <b>Candidate B</b><br>$m = \frac{1}{\sqrt{3}}$ (with or without a diagram) • <sup>1</sup> ^ • <sup>2</sup> ✓ 2<br>$y = \frac{1}{\sqrt{3}}x + \frac{2}{\sqrt{3}}$<br><br>• <sup>3</sup> ✓ 1                                                                                                                     |          |
| <b>Candidate C</b><br>$m = \tan \theta$ (with or without a diagram)<br>$m = \frac{1}{\sqrt{3}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  | <ul style="list-style-type: none"><li>•<sup>1</sup> ^</li><li>•<sup>2</sup> ✓ 1</li></ul>                                                                                          | <b>Candidate D</b><br>$m = \tan \theta$ (with or without a diagram)<br>$m = \sqrt{3}$<br>$y = \sqrt{3}x + 2\sqrt{3}$<br><br>• <sup>1</sup> ^<br>• <sup>2</sup> ✗<br>• <sup>3</sup> ✓ 1                                                                                                                         |          |
| <b>Candidate E</b><br>$m = \tan \theta = \frac{\pi}{6}$<br>$m = \frac{1}{\sqrt{3}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |  | <ul style="list-style-type: none"><li>•<sup>1</sup> ✗</li><li>•<sup>2</sup> ✓ 1</li></ul>                                                                                          | <b>Candidate F</b><br>$m = \tan \frac{\pi}{3}$<br>$m = 60$<br>$y = 60(x + 2)$<br><br>• <sup>1</sup> ✗<br>• <sup>2</sup> ✗<br>• <sup>3</sup> ✗                                                                                                                                                                  |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |  | Generic Scheme                                                                                                                                                                                                       | Illustrative Scheme                                                                                                                                                                                                                                                                                                                             | Max Mark |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 6.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> start to integrate</li> <li>•<sup>2</sup> complete integration</li> <li>•<sup>3</sup> process limits</li> <li>•<sup>4</sup> evaluate integral</li> </ul>        | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\frac{(10-3x)^{\frac{1}{2}}}{\frac{1}{2}} \dots</math></li> <li>•<sup>2</sup> <math>\dots \times -\frac{1}{3}</math></li> <li>•<sup>3</sup> <math>-\frac{2}{3}(10-3(2))^{\frac{1}{2}} - \left(-\frac{2}{3}(10-3(-5))^{\frac{1}{2}}\right)</math></li> <li>•<sup>4</sup> 2</li> </ul> | 4        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                 |          |
| <ol style="list-style-type: none"> <li>For candidates who <b>differentiate throughout</b> or make <b>no attempt to integrate</b>, award 0/4.</li> <li>If candidates start to integrate individual terms within the bracket or attempt to expand a bracket or use another invalid approach no further marks are available.</li> <li>Do not penalise the inclusion of '+c' or the continued appearance of the integral sign after •<sup>1</sup>.</li> <li>•<sup>3</sup> is only available for substitution into an expression which is equivalent to the integrand obtained at •<sup>2</sup>.</li> <li>The integral obtained must contain a non-integer power for •<sup>4</sup> to be available.</li> <li>•<sup>4</sup> is only available to candidates who deal with the coefficient of x at the •<sup>2</sup> stage. See Candidate A.</li> </ol> |  |  |                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                 |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                 |          |
| Candidate A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  | Candidate B - NOT differentiating throughout                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                 |          |
| $\frac{(10-3x)^{\frac{1}{2}}}{\frac{1}{2}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  | $-\frac{1}{2}(10-3x)^{-\frac{3}{2}} \times -\frac{1}{3}$                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                 |          |
| $2(10-3(2))^{\frac{1}{2}} - 2(10-3(-5))^{\frac{1}{2}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  | $\frac{1}{6}(10-3(2))^{-\frac{3}{2}} - \frac{1}{6}(10-3(-5))^{-\frac{3}{2}}$                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                 |          |
| -6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |  | $\frac{39}{2000}$                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                 |          |
| <ul style="list-style-type: none"> <li>•<sup>1</sup> ✓</li> <li>•<sup>2</sup> ✗</li> <li>•<sup>3</sup> <input checked="" type="checkbox"/> 1</li> <li>•<sup>4</sup> <input checked="" type="checkbox"/> 2</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> ✗</li> <li>•<sup>2</sup> ✓</li> <li>•<sup>3</sup> <input checked="" type="checkbox"/> 1</li> <li>•<sup>4</sup> <input checked="" type="checkbox"/> 1</li> </ul> |                                                                                                                                                                                                                                                                                                                                                 |          |
| Candidate C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  | Candidate D - integrating over two lines                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                 |          |
| $\frac{(10-3x)^{\frac{1}{2}}}{\frac{1}{2}} \times -3$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |  | $\frac{(10-3x)^{\frac{1}{2}}}{\frac{1}{2}}$                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                 |          |
| $-6(10-3(2))^{\frac{1}{2}} - \left(-6(10-3(-5))^{\frac{1}{2}}\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |  | $\frac{(10-3x)^{\frac{1}{2}}}{\frac{1}{2}} \times -\frac{1}{3}$                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                 |          |
| 18                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |  | $-\frac{2}{3}(10-3(2))^{\frac{1}{2}} - \left(-\frac{2}{3}(10-3(-5))^{\frac{1}{2}}\right)$                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                 |          |
| <ul style="list-style-type: none"> <li>•<sup>1</sup> ✓</li> <li>•<sup>2</sup> ✗</li> <li>•<sup>3</sup> <input checked="" type="checkbox"/> 1</li> <li>•<sup>4</sup> <input checked="" type="checkbox"/> 1</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> ✓</li> <li>•<sup>2</sup> ✗</li> <li>•<sup>3</sup> <input checked="" type="checkbox"/> 1</li> <li>•<sup>4</sup> <input checked="" type="checkbox"/> 1</li> </ul> |                                                                                                                                                                                                                                                                                                                                                 |          |

| Question                                                                                   |     |      | Generic Scheme                                                               | Illustrative Scheme                                                                                                                                                                                                                                                                                                | Max Mark |
|--------------------------------------------------------------------------------------------|-----|------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7.                                                                                         | (a) | (i)  | • <sup>1</sup> determine $\sin r$                                            | • <sup>1</sup> $\frac{1}{\sqrt{10}}$                                                                                                                                                                                                                                                                               | 1        |
|                                                                                            |     | (ii) | • <sup>2</sup> determine $\sin q$                                            | • <sup>2</sup> $\frac{3}{\sqrt{13}}$                                                                                                                                                                                                                                                                               | 1        |
| <b>Notes:</b>                                                                              |     |      |                                                                              |                                                                                                                                                                                                                                                                                                                    |          |
| 1. In (a)(ii), where candidates do not simplify the perfect square see Candidates A and B. |     |      |                                                                              |                                                                                                                                                                                                                                                                                                                    |          |
| <b>Commonly Observed Responses:</b>                                                        |     |      |                                                                              |                                                                                                                                                                                                                                                                                                                    |          |
| <b>Candidate A</b><br>$\sin q = \frac{\sqrt{9}}{\sqrt{13}}$                                |     |      | • <sup>2</sup> <span style="border: 1px solid red; padding: 2px;">✓ 2</span> | <b>Candidate B - simplification in part (b)</b><br>(a)(ii) $\sin q = \frac{\sqrt{9}}{\sqrt{13}}$ • <sup>2</sup> ✓<br>(b) $\sin(q-r) = \frac{7}{\dots}$ <div style="border: 1px solid black; border-radius: 10px; padding: 5px; display: inline-block; margin-top: 10px;"> Roots have been simplified in (b) </div> |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |  | Generic Scheme                                                                                                                                                                                                       | Illustrative Scheme                                                                                                                                                                                                                                                                         | Max Mark |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | (b) |  | <p>•<sup>3</sup> select appropriate formula and express in terms of <math>p</math> and <math>q</math></p> <p>•<sup>4</sup> substitute into addition formula</p> <p>•<sup>5</sup> evaluate <math>\sin(q-r)</math></p> | <p>•<sup>3</sup> <math>\sin q \cos r - \cos q \sin r</math> stated or implied by •<sup>4</sup></p> <p>•<sup>4</sup> <math>\frac{3}{\sqrt{13}} \times \frac{3}{\sqrt{10}} - \frac{2}{\sqrt{13}} \times \frac{1}{\sqrt{10}}</math></p> <p>•<sup>5</sup> <math>\frac{7}{\sqrt{130}}</math></p> | 3        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |  |                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                             |          |
| <p>2. Award •<sup>3</sup> for candidates who write <math>\sin\left(\frac{3}{\sqrt{13}}\right) \times \cos\left(\frac{3}{\sqrt{10}}\right) - \sin\left(\frac{2}{\sqrt{13}}\right) \times \cos\left(\frac{1}{\sqrt{10}}\right)</math>. •<sup>4</sup> and •<sup>5</sup> are unavailable.</p> <p>3. For any attempt to use <math>\sin(q-r) = \sin q - \sin r</math>, •<sup>4</sup> and •<sup>5</sup> are unavailable.</p> <p>4. At •<sup>5</sup>, the answer must be given as a single fraction. Accept <math>\frac{7}{\sqrt{13}\sqrt{10}}</math>, <math>\frac{7\sqrt{10}}{10\sqrt{13}}</math> and <math>\frac{7\sqrt{13}}{13\sqrt{10}}</math>.</p> <p>5. Do not penalise trigonometric ratios which are less than <math>-1</math> or greater than <math>1</math>.</p> |     |  |                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                             |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |  |                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                             |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |  |                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                             |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  | Generic Scheme                                                                                                                                                                                                                                                                                                                      | Illustrative Scheme                                                                                                                                                                                                                                                                                                | Max Mark |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 8.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  | <b>Method 1</b> <ul style="list-style-type: none"><li>•<sup>1</sup> apply <math>\log_6 x + \log_6 y = \log_6 xy</math></li><li>•<sup>2</sup> write in exponential form</li><li>•<sup>3</sup> express in standard quadratic form</li><li>•<sup>4</sup> solve quadratic and state solution of log equation</li></ul>                  | <b>Method 1</b> <ul style="list-style-type: none"><li>•<sup>1</sup> <math>\log_6 (x(x+5)) = \dots</math></li><li>•<sup>2</sup> <math>x(x+5) = 6^2</math></li><li>•<sup>3</sup> <math>x^2 + 5x - 36 = 0</math></li><li>•<sup>4</sup> <math>-9, 4</math> and <math>x &gt; 0 \Rightarrow x = 4</math></li></ul>       | 4        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  | <b>Method 2</b> <ul style="list-style-type: none"><li>•<sup>1</sup> apply <math>\log_6 x + \log_6 y = \log_6 xy</math></li><li>•<sup>2</sup> apply <math>m \log_6 x = \log_6 x^m</math></li><li>•<sup>3</sup> express in standard quadratic form</li><li>•<sup>4</sup> solve quadratic and state solution of log equation</li></ul> | <b>Method 2</b> <ul style="list-style-type: none"><li>•<sup>1</sup> <math>\log_6 (x(x+5)) = \dots</math></li><li>•<sup>2</sup> <math>\dots = \log_6 6^2</math></li><li>•<sup>3</sup> <math>x^2 + 5x - 36 = 0</math></li><li>•<sup>4</sup> <math>-9, 4</math> and <math>x &gt; 0 \Rightarrow x = 4</math></li></ul> |          |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  |                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                    |          |
| <p>1. Accept <math>\log_6 x(x+5) = \dots</math> for •<sup>1</sup>.</p> <p>2. •<sup>2</sup> is not available for <math>x(x+5) = 2^6</math>; however candidates may still gain •<sup>3</sup> and •<sup>4</sup>.</p> <p>3. •<sup>3</sup> and •<sup>4</sup> are only available if the quadratic reached at •<sup>3</sup> is obtained by applying the rules in •<sup>1</sup> and •<sup>2</sup>.</p> <p>4. •<sup>4</sup> is only available for solving a polynomial of degree two or higher.</p> <p>5. At •<sup>4</sup>, accept any indication that <math>-9</math> has been discarded. For example, scoring out <math>x = -9</math> or underlining <math>x = 4</math>.</p> |  |  |                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                    |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |  |                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                    |          |
| <b>Candidate A</b><br>$\log_6 (x(x+5)) = 2$<br>$x(x+5) = 12$<br>$x^2 + 5x - 12 = 0$<br>$\frac{-5 \pm \sqrt{73}}{2}$ and $x > 0 \Rightarrow x = \frac{-5 + \sqrt{73}}{2}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  | • <sup>1</sup> ✓<br>• <sup>2</sup> ✗<br>• <sup>3</sup> ✓ 1<br>• <sup>4</sup> ✓ 1                                                                                                                                                                                                                                                    | <b>Candidate B</b><br>$\log_6 (x(x+5)) = 2$<br>$x(x+5) = 64$<br>$x^2 + 5x - 64 = 0$<br>$\frac{-5 \pm \sqrt{281}}{2}$ and $x > 0 \Rightarrow x = \frac{-5 + \sqrt{281}}{2}$                                                                                                                                         |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                                                                                                                                                                                                                                                                                                                                     | • <sup>1</sup> ✓<br>• <sup>2</sup> ✗<br>• <sup>3</sup> ✓ 1<br>• <sup>4</sup> ✓ 1                                                                                                                                                                                                                                   |          |

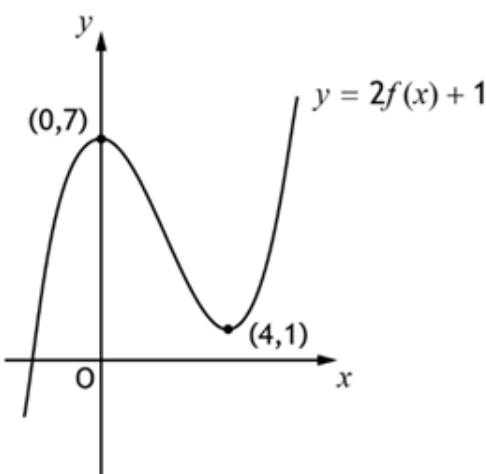
| Question |  |  | Generic Scheme                                                                                                                                                                                                                                                                                                                | Illustrative Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Max Mar |
|----------|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 9.       |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> substitute for <math>\cos 2x^\circ</math> into equation</li> <li>•<sup>2</sup> express in standard quadratic form</li> <li>•<sup>3</sup> factorise</li> <li>•<sup>4</sup> solve for <math>\cos x^\circ</math></li> <li>•<sup>5</sup> solve for <math>x</math></li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>2 \cos^2 x^\circ - 1 \dots</math></li> <li>•<sup>2</sup> <math>2 \cos^2 x^\circ - 5 \cos x^\circ + 2 = 0</math></li> <li>•<sup>3</sup> <math>(2 \cos x^\circ - 1)(\cos x^\circ - 2) = 0</math></li> <li style="text-align: center;">•<sup>4</sup>                      •<sup>5</sup></li> <li>•<sup>4</sup> <math>\cos x^\circ = \frac{1}{2}</math>      <math>\cos x^\circ = 2</math></li> <li>•<sup>5</sup> <math>x = 60, 300</math>      'no solutions'</li> </ul> | 5       |

#### Notes:

1. •<sup>1</sup> is not available for simply stating  $\cos 2x^\circ = 2 \cos^2 x^\circ - 1$  with no further working.
2. In the event of  $\cos^2 x^\circ - \sin^2 x^\circ$  or  $1 - 2 \sin^2 x^\circ$  being substituted for  $\cos 2x^\circ$ , •<sup>1</sup> cannot be awarded until the equation reduces to a quadratic in  $\cos x^\circ$ .
3. Substituting  $2 \cos^2 A - 1$  or  $2 \cos^2 \alpha - 1$  for  $\cos 2x^\circ$  at the •<sup>1</sup> stage should be treated as bad form provided the equation is written in terms of  $x$  at •<sup>2</sup> stage. Otherwise, •<sup>1</sup> is not available.
4. Do not penalise the omission of degree signs.
5. '= 0' must appear by •<sup>3</sup> stage for •<sup>2</sup> to be awarded. However, for candidates using the quadratic formula to solve the equation, '= 0' must appear at •<sup>2</sup> stage for •<sup>2</sup> to be awarded.
6.  $\cos x^\circ = \frac{5 \pm \sqrt{9}}{4}$  gains •<sup>3</sup>.
7. Candidates may express the equation obtained at •<sup>2</sup> in the form  $2c^2 - 5c + 2 = 0$  or  $2x^2 - 5x + 2 = 0$ . In these cases, award •<sup>3</sup> for  $(2c - 1)(c - 2) = 0$  or  $(2x - 1)(x - 2) = 0$ . However, •<sup>4</sup> is only available if  $\cos x^\circ$  appears explicitly at this stage. See Candidate A.
8. The equation  $2 + 2 \cos^2 x^\circ - 5 \cos x^\circ = 0$  does not gain •<sup>2</sup> unless •<sup>3</sup> has been awarded.
9. •<sup>4</sup> and •<sup>5</sup> are only available as a consequence of trying to solve a quadratic equation. See Candidate B. However, •<sup>5</sup> is not available if the quadratic equation has repeated roots.
10. •<sup>3</sup>, •<sup>4</sup> and •<sup>5</sup> are not available for any attempt to solve a quadratic equation written in the form  $ax^2 + bx = c$ . See Candidate C.
11. •<sup>5</sup> is only available for 2 valid solutions within the stated range. Ignore 'solutions' outwith the range. However, see Candidate E.
12. Accept  ~~$\cos x^\circ = 2$~~  for •<sup>5</sup>. See Candidate A.

| Question                                                                       |             | Generic Scheme |                                                                                                 | Illustrative Scheme |  | Max Mark |
|--------------------------------------------------------------------------------|-------------|----------------|-------------------------------------------------------------------------------------------------|---------------------|--|----------|
| 9.                                                                             | (continued) |                |                                                                                                 |                     |  |          |
| Commonly Observed Responses:                                                   |             |                |                                                                                                 |                     |  |          |
| <b>Candidate A</b>                                                             |             |                | <b>Candidate B - not solving a quadratic</b>                                                    |                     |  |          |
| $2 \cos^2 x^\circ - 1 = 5 \cos x^\circ - 3$                                    |             |                | $2 \cos^2 x^\circ - 1 = 5 \cos x^\circ - 3$                                                     |                     |  |          |
| $2c^2 - 5c + 2 = 0$                                                            |             |                | $2 \cos^2 x^\circ - 5 \cos x^\circ + 2 = 0$                                                     |                     |  |          |
| $(2c - 1)(c - 2) = 0$                                                          |             |                | $-3 \cos x^\circ + 2 = 0$                                                                       |                     |  |          |
| $c = \frac{1}{2}, c = 2$                                                       |             |                | $\cos x^\circ = \frac{2}{3}$                                                                    |                     |  |          |
| $x = 60, 300$ <del><math>\cos x^\circ = 2</math></del>                         |             |                |                                                                                                 |                     |  |          |
| • <sup>1</sup> ✓                                                               |             |                | • <sup>1</sup> ✓                                                                                |                     |  |          |
| • <sup>2</sup> ✓                                                               |             |                | • <sup>2</sup> ✓                                                                                |                     |  |          |
| • <sup>3</sup> ✓                                                               |             |                | • <sup>3</sup> ✗                                                                                |                     |  |          |
| • <sup>4</sup> ✗                                                               |             |                | • <sup>4</sup> <span style="border: 1px solid red; padding: 0 2px;">✓ 2</span> • <sup>5</sup> ^ |                     |  |          |
| • <sup>5</sup> <span style="border: 1px solid red; padding: 0 2px;">✓ 1</span> |             |                |                                                                                                 |                     |  |          |
| <b>Candidate C - not in standard quadratic form</b>                            |             |                | <b>Candidate D - reading <math>\cos 2x^\circ</math> as <math>\cos^2 x^\circ</math></b>          |                     |  |          |
| $2 \cos^2 x^\circ - 1 = 5 \cos x^\circ - 3$                                    |             |                | $\cos^2 x^\circ = 5 \cos x^\circ - 3$                                                           |                     |  |          |
| $2 \cos^2 x^\circ - 5 \cos x^\circ = -2$                                       |             |                | $\cos^2 x^\circ - 5 \cos x^\circ + 3 = 0$                                                       |                     |  |          |
| $\cos x^\circ (2 \cos x^\circ - 5) = -2$                                       |             |                | $\cos x^\circ = \frac{5 \pm \sqrt{13}}{2}$                                                      |                     |  |          |
| $\cos x^\circ = -2, 2 \cos x^\circ - 5 = -2$                                   |             |                |                                                                                                 |                     |  |          |
| $\Rightarrow \cos x = \frac{3}{2}$                                             |             |                |                                                                                                 |                     |  |          |
| No solutions                                                                   |             |                |                                                                                                 |                     |  |          |
| • <sup>1</sup> ✓                                                               |             |                | • <sup>1</sup> ✗                                                                                |                     |  |          |
| • <sup>2</sup> <span style="border: 1px solid red; padding: 0 2px;">✓ 2</span> |             |                | • <sup>2</sup> <span style="border: 1px solid red; padding: 0 2px;">✓ 1</span>                  |                     |  |          |
| • <sup>3</sup> <span style="border: 1px solid red; padding: 0 2px;">✓ 2</span> |             |                | • <sup>3</sup> <span style="border: 1px solid red; padding: 0 2px;">✓ 1</span>                  |                     |  |          |
| • <sup>4</sup> ✗                                                               |             |                | • <sup>4</sup> ^ • <sup>5</sup> ^                                                               |                     |  |          |
| • <sup>5</sup> ✗                                                               |             |                |                                                                                                 |                     |  |          |
| <b>Candidate E</b>                                                             |             |                |                                                                                                 |                     |  |          |
| • <sup>1</sup> ✓ • <sup>2</sup> ✓                                              |             |                |                                                                                                 |                     |  |          |
| $(\cos x^\circ - 1)(\cos x^\circ - 2) = 0$                                     |             |                |                                                                                                 |                     |  |          |
| $\cos x^\circ = 1, \cos x^\circ = 2$                                           |             |                |                                                                                                 |                     |  |          |
| $x = 0$ No solutions                                                           |             |                |                                                                                                 |                     |  |          |
| • <sup>3</sup> ✗                                                               |             |                |                                                                                                 |                     |  |          |
| • <sup>4</sup> <span style="border: 1px solid red; padding: 0 2px;">✓ 1</span> |             |                |                                                                                                 |                     |  |          |
| • <sup>5</sup> <span style="border: 1px solid red; padding: 0 2px;">✓ 1</span> |             |                |                                                                                                 |                     |  |          |



| Question |     |  | Generic Scheme                                                                                                                                                                                                                                                              | Illustrative Scheme                                                                                                                                                                              | Max Mark |
|----------|-----|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10.      | (a) |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> vertical scaling by a factor of 2 identifiable from graph</li> <li>•<sup>2</sup> vertical translation of '+1' units identifiable from graph</li> <li>•<sup>3</sup> transformations applied in correct order</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup></li> <li>•<sup>2</sup></li> <li>•<sup>3</sup></li> </ul>  | 3        |

#### Notes:

1. •<sup>1</sup>, •<sup>2</sup> and •<sup>3</sup> are only available for a 'cubic' with a maximum and minimum turning point.
2. Ignore intersections (or lack of intersections) with the original graph.

#### Commonly Observed Responses:

Where the image of (4,0) is not (4,1), that point must be annotated (or drawn to within tolerance). In the following table, the images of the given points must be stationary points for the marks to be awarded.

| Image of (0,3) | Image of (4,0)      | Award... |                                                                  |
|----------------|---------------------|----------|------------------------------------------------------------------|
| (0,8)          | (4,2)               | 2/3      | Transformation in wrong order                                    |
| (0,4)          | (8,1)               | 1/3      | Only vertical translation correct                                |
| (0,4)          | (4,1)               | 1/3      |                                                                  |
| (0,4)          | (2,1)               | 1/3      |                                                                  |
| (0,5)          | (4,-1)              | 2/3      | Evidence of vertical scaling and transformation in correct order |
| (0,6)          | (4,0)               | 1/3      | Evidence of vertical scaling                                     |
| (0,7)          | any incorrect point | 1/3      |                                                                  |
| (1,6)          | (5,0)               | 1/3      |                                                                  |
| (-1,6)         | (3,0)               | 1/3      |                                                                  |
| (0,-2)         | (4,1)               | 1/3      | Evidence of vertical translation                                 |
| (0,4)          | (-4,1)              | 1/3      |                                                                  |
| (0,5)          | any other point     | 0/3      | Insufficient evidence of scaling/translation                     |
| (0,2)          | any other point     | 0/3      |                                                                  |

| Question                     |     |  | Generic Scheme                                                                                         | Illustrative Scheme                                                             | Max Mark |
|------------------------------|-----|--|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------|
| 10.                          | (b) |  | <ul style="list-style-type: none"> <li>•<sup>4</sup> state coordinates of stationary points</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>4</sup> (0,3) and (8,0)</li> </ul> | 1        |
| Notes:                       |     |  |                                                                                                        |                                                                                 |          |
|                              |     |  |                                                                                                        |                                                                                 |          |
| Commonly Observed Responses: |     |  |                                                                                                        |                                                                                 |          |
|                              |     |  |                                                                                                        |                                                                                 |          |

| Question                                                                                                                                                                               |  |  | Generic Scheme                                                                                                                                                                                                                                                | Illustrative Scheme                                                                                                                                                                                                                                  | Max Mark |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 11.                                                                                                                                                                                    |  |  | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> identify common factor</li> <li>•<sup>2</sup> complete the square</li> <li>•<sup>3</sup> process for <math>r</math> and write in required form</li> </ul>                                | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>2(x^2 + 6x \dots</math> stated or implied by •<sup>2</sup></li> <li>•<sup>2</sup> <math>2(x+3)^2 \dots</math></li> <li>•<sup>3</sup> <math>2(x+3)^2 + 5</math></li> </ul> | 3        |
|                                                                                                                                                                                        |  |  | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> expand completed square form</li> <li>•<sup>2</sup> equate coefficients</li> <li>•<sup>3</sup> process for <math>q</math> and <math>r</math> and write in required form</li> </ul>       | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>px^2 + 2pqx + pq^2 + r</math></li> <li>•<sup>2</sup> <math>p = 2, 2pq = 12, pq^2 + r = 23</math></li> <li>•<sup>3</sup> <math>2(x+3)^2 + 5</math></li> </ul>              |          |
| <b>Notes:</b>                                                                                                                                                                          |  |  |                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                      |          |
| 1. $2(x+3)^2 + 5$ with no working gains • <sup>1</sup> and • <sup>2</sup> only. However, see Candidate E.                                                                              |  |  |                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                      |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                    |  |  |                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                      |          |
| <b>Candidate A</b><br>$2(x^2 + 6) + 23$<br>$2((x+3)^2 - 9) + 23$ • <sup>1</sup> ✓    • <sup>2</sup> ✓<br>$2(x+3)^2 + 5$ • <sup>3</sup> ✓<br>See the exception to marking principle (h) |  |  | <b>Candidate B</b><br>$px^2 + 2pqx + pq^2 + r$ • <sup>1</sup> ✓<br>$p = 2, 2pq = 12, pq^2 + r = 23$ • <sup>2</sup> ✓<br>$q = 3, r = 5$ • <sup>3</sup> ^<br><div>             •<sup>3</sup> is lost as answer is not in completed square form           </div> |                                                                                                                                                                                                                                                      |          |
| <b>Candidate C</b><br>$2(x^2 + 12x) + 23$ • <sup>1</sup> ✗<br>$2((x+6)^2 - 36) + 23$ • <sup>2</sup> ✓ 1<br>$2(x+6)^2 - 49$ • <sup>3</sup> ✓ 1                                          |  |  | <b>Candidate D</b><br>$2((x+6)^2 - 36) + 23$ • <sup>1</sup> ✗    • <sup>2</sup> ✗<br>$2(x+6)^2 - 49$ • <sup>3</sup> ✓ 1                                                                                                                                       |                                                                                                                                                                                                                                                      |          |
| <b>Candidate E</b><br>$2(x+3)^2 + 5$ • <sup>1</sup> ✓    • <sup>2</sup> ✓<br>Check: $= 2(x^2 + 6x + 9) + 5$<br>$= 2x^2 + 12x + 18 + 5$<br>$= 2x^2 + 12x + 23$ • <sup>3</sup> ✓         |  |  |                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                      |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  | Generic Scheme                                                                                                                                                                    | Illustrative Scheme                                                                                                                                                                                                       | Max Mark                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> start to differentiate</li> <li>•<sup>2</sup> complete differentiation</li> <li>•<sup>3</sup> evaluate derivative</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>4 \cos\left(3x - \frac{\pi}{3}\right) \dots</math></li> <li>•<sup>2</sup> <math>\dots \times 3</math></li> <li>•<sup>3</sup> <math>6\sqrt{3}</math></li> </ul> | 3                                                                                                                                                    |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |                                                                                                                                                                                   |                                                                                                                                                                                                                           |                                                                                                                                                      |
| <ol style="list-style-type: none"> <li>Where candidates make no attempt to differentiate or use another invalid approach, •<sup>2</sup> and •<sup>3</sup> are not available.</li> <li>At the •<sup>1</sup> and •<sup>2</sup> stage, candidates who work in degrees cannot gain •<sup>1</sup>. However •<sup>2</sup> and •<sup>3</sup> are still available.</li> <li>At the •<sup>3</sup> stage, do not penalise candidates who work in degrees or in radians and degrees.</li> <li>Ignore the appearance of +c at any stage.</li> </ol> |  |  |                                                                                                                                                                                   |                                                                                                                                                                                                                           |                                                                                                                                                      |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |                                                                                                                                                                                   |                                                                                                                                                                                                                           |                                                                                                                                                      |
| <b>Candidate A</b><br>Differentiating over two lines<br>$f'(x) = 4 \cos\left(3x - \frac{\pi}{3}\right)$ • <sup>1</sup> ✓<br>$f'(x) = 12 \cos\left(3x - \frac{\pi}{3}\right)$ • <sup>2</sup> ^<br>$6\sqrt{3}$ • <sup>3</sup> ✓ 1                                                                                                                                                                                                                                                                                                         |  |  | <b>Candidate B</b><br>$4 \cos\left(3x - \frac{\pi}{3}\right) \times \frac{1}{3}$ • <sup>1</sup> ✓ • <sup>2</sup> ✗<br>$\frac{2\sqrt{3}}{3}$ • <sup>3</sup> ✓ 1                    |                                                                                                                                                                                                                           | <b>Candidate C</b><br>$4 \cos\left(3x - \frac{\pi}{3}\right)$ • <sup>1</sup> ✓ • <sup>2</sup> ^<br>$2\sqrt{3}$ • <sup>3</sup> ✓ 1                    |
| <b>Candidate D</b><br>$\pm 12 \sin\left(3x - \frac{\pi}{3}\right)$ • <sup>1</sup> ✗<br>$\pm 6$ • <sup>2</sup> ✗<br>$\pm 6$ • <sup>3</sup> ✓ 1                                                                                                                                                                                                                                                                                                                                                                                           |  |  | <b>Candidate E</b><br>$\pm 4 \sin\left(3x - \frac{\pi}{3}\right) \dots$ • <sup>1</sup> ✗<br>$\dots \times 3$ • <sup>2</sup> ✓ 1<br>$\pm 6$ • <sup>3</sup> ✓ 1                     |                                                                                                                                                                                                                           | <b>Candidate F</b><br>$-12 \cos\left(3x - \frac{\pi}{3}\right)$ • <sup>1</sup> ✗<br>$-6\sqrt{3}$ • <sup>2</sup> ✓<br>$-6\sqrt{3}$ • <sup>3</sup> ✓ 1 |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |      | Generic Scheme                                                                                                                                                                                                | Illustrative Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Max Mark |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 13.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | (a) | (i)  | <ul style="list-style-type: none"> <li>•<sup>1</sup> use <math>-2</math> in synthetic division or evaluation of the cubic</li> <li>•<sup>2</sup> complete division/evaluation and interpret result</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math display="block">\begin{array}{r rrrr} -2 &amp; 1 &amp; -2 &amp; -20 &amp; -24 \\ &amp; &amp; 1 &amp; &amp; \\ \hline &amp; &amp; &amp; &amp; \end{array}</math> or<br/> <math>(-2)^3 - 2(-2)^2 - 20(-2) - 24</math> </li> <li>•<sup>2</sup> <math display="block">\begin{array}{r rrrr} -2 &amp; 1 &amp; -2 &amp; -20 &amp; -24 \\ &amp; &amp; -2 &amp; 8 &amp; 24 \\ \hline &amp; 1 &amp; -4 &amp; -12 &amp; 0 \end{array}</math> Remainder = 0 <math>\therefore (x+2)</math> is a factor<br/> or<br/> <math>f(-2) = 0 \therefore (x+2)</math> is a factor </li> </ul> | 2        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     | (ii) | <ul style="list-style-type: none"> <li>•<sup>3</sup> state quadratic factor</li> <li>•<sup>4</sup> find remaining factors or apply the quadratic formula</li> <li>•<sup>5</sup> state solution</li> </ul>     | <ul style="list-style-type: none"> <li>•<sup>3</sup> <math>x^2 - 4x - 12</math></li> <li>•<sup>4</sup> <math>(x+2)</math> and <math>(x-6)</math><br/>or<br/> <math display="block">\frac{4 \pm \sqrt{(-4)^2 - 4(1)(-12)}}{2(1)}</math> </li> <li>•<sup>5</sup> <math>-2, 6</math></li> </ul>                                                                                                                                                                                                                                                                                                                                       | 3        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |      |                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |
| 1. Communication at • <sup>2</sup> must be consistent with working at that stage - a candidate's working must arrive legitimately at 0 before • <sup>2</sup> can be awarded.<br>2. Accept any of the following for • <sup>2</sup> : <ul style="list-style-type: none"> <li>• ' <math>f(-2) = 0</math> so <math>(x+2)</math> is a factor'</li> <li>• 'since remainder = 0, it is a factor'</li> <li>• the '0' from any method linked to the word 'factor' by 'so', 'hence', <math>\therefore</math>, <math>\rightarrow</math>, <math>\Rightarrow</math> etc.</li> </ul> 3. Do not accept any of the following for • <sup>2</sup> : <ul style="list-style-type: none"> <li>• double underlining the '0' or boxing the '0' without comment</li> <li>• '<math>x = -2</math> is a factor', '<math>\dots</math> is a root'</li> <li>• the word 'factor' only, with no link.</li> </ul> |     |      |                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |      |                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | (b) |      | • <sup>6</sup> state value of $k$                                                                                                                                                                             | • <sup>6</sup> 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |      |                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |
| 1. Accept $y = f(x-3)$ or $f(x-3)$ for • <sup>6</sup> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |      |                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |      |                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |

| Question                                                                                                           |     |      | Generic Scheme                                                                                                                                                          | Illustrative Scheme                                                                                                                                                                  | Max Mark |
|--------------------------------------------------------------------------------------------------------------------|-----|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 14.                                                                                                                | (a) | (i)  | <ul style="list-style-type: none"> <li>•<sup>1</sup> state coordinates of centre</li> <li>•<sup>2</sup> state radius</li> </ul>                                         | <ul style="list-style-type: none"> <li>•<sup>1</sup> (7, -5)</li> <li>•<sup>2</sup> 10</li> </ul>                                                                                    | 2        |
|                                                                                                                    |     | (ii) | <ul style="list-style-type: none"> <li>•<sup>3</sup> substitute coordinates of P and evaluate</li> <li>•<sup>4</sup> communicate result</li> </ul>                      | <ul style="list-style-type: none"> <li>•<sup>3</sup> <math>(-2 - 7)^2 + (7 + 5)^2 = 225</math></li> <li>•<sup>4</sup> <math>225 &gt; 100 \therefore P</math> lies outside</li> </ul> | 2        |
| <b>Notes:</b>                                                                                                      |     |      |                                                                                                                                                                         |                                                                                                                                                                                      |          |
| 1. Accept $x = 7, y = -5$ for • <sup>1</sup> .<br>2. Do not accept $g = 7, f = -5$ or $7, -5$ for • <sup>1</sup> . |     |      |                                                                                                                                                                         |                                                                                                                                                                                      |          |
| <b>Commonly Observed Responses:</b>                                                                                |     |      |                                                                                                                                                                         |                                                                                                                                                                                      |          |
| <b>Candidate A</b>                                                                                                 |     |      | $d = 15$ • <sup>3</sup> ✓<br>$15 > 10 \therefore P$ lies outside • <sup>4</sup> ✓                                                                                       | <b>Candidate B</b><br>$d = \sqrt{225}$ • <sup>3</sup> ✓<br>$\sqrt{225} > 10 \therefore P$ lies outside • <sup>4</sup> ✓                                                              |          |
| <b>Candidate C</b>                                                                                                 |     |      | $d = 15$ • <sup>3</sup> ✓<br>$r = 10$<br>$d > r \therefore P$ lies outside • <sup>4</sup> ✓                                                                             |                                                                                                                                                                                      |          |
|                                                                                                                    | (b) |      | <ul style="list-style-type: none"> <li>•<sup>5</sup> determine first value of <math>r</math></li> <li>•<sup>6</sup> determine second value of <math>r</math></li> </ul> | <ul style="list-style-type: none"> <li>•<sup>5</sup> 5</li> <li>•<sup>6</sup> 25</li> </ul>                                                                                          | 2        |
| <b>Notes:</b>                                                                                                      |     |      |                                                                                                                                                                         |                                                                                                                                                                                      |          |
|                                                                                                                    |     |      |                                                                                                                                                                         |                                                                                                                                                                                      |          |
| <b>Commonly Observed Responses:</b>                                                                                |     |      |                                                                                                                                                                         |                                                                                                                                                                                      |          |
|                                                                                                                    |     |      |                                                                                                                                                                         |                                                                                                                                                                                      |          |

[END OF MARKING INSTRUCTIONS]